

Report 11/05/2020

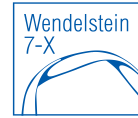
P. Hacker

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QSB LabVIEW Routine Review (see CHANGELOG.md)



PROS

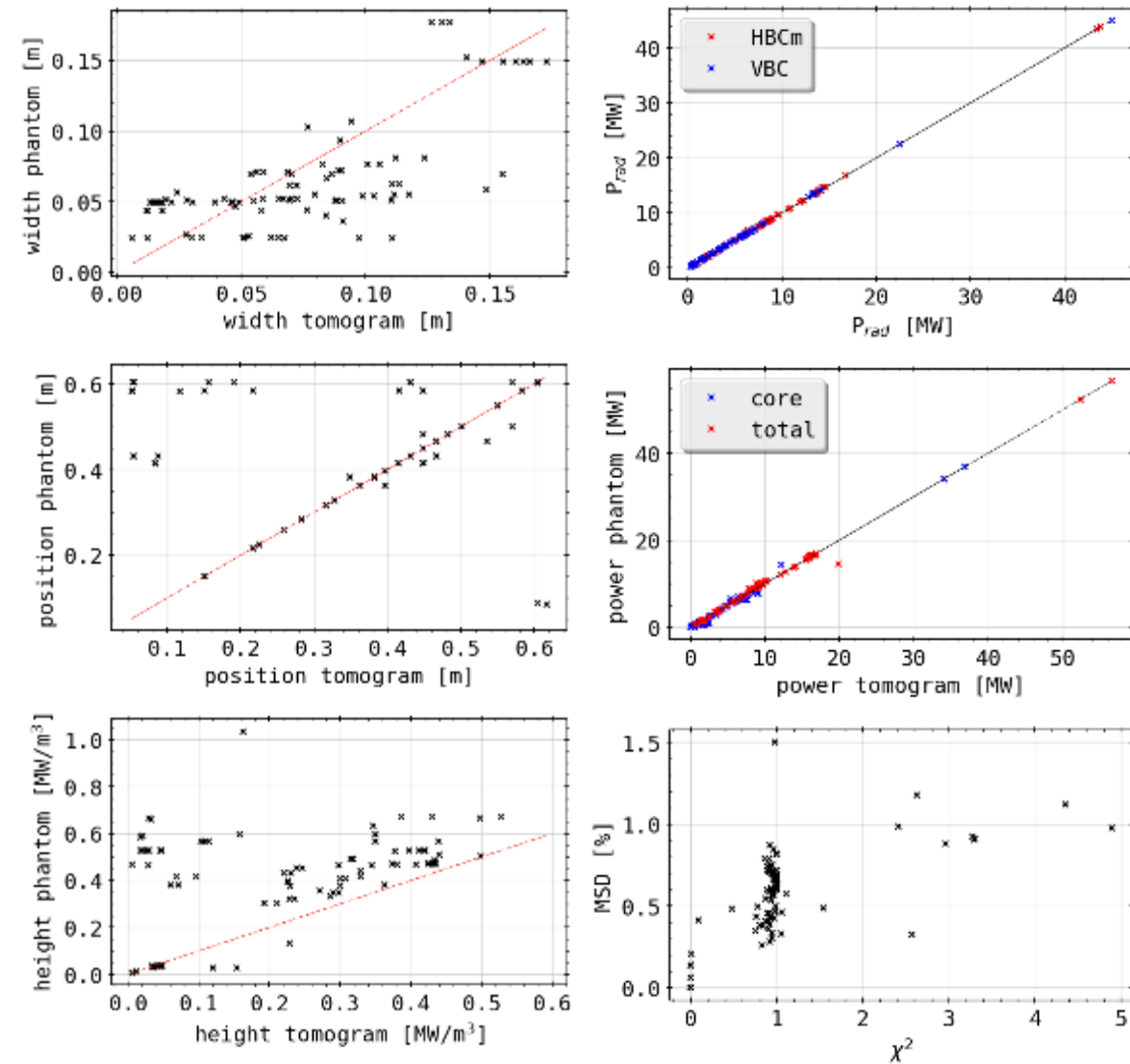
- added EPOCH timer to UI, timing mode that calculates EPOCH time at start
- added T1-T4 timing input to UI, added T0-T1 timer to UI (should at least be 30s)
- fixed wrong time base in calibration procedure
- DAC range setting file load or input via the UI
- **selected channel feedback output**, though with delay >10ms (single/multi channel)
- performance for 1-10 selected channels in feedback is fine
- overhaul of calibration parameter routines since detector response is more complex than a single peak and exponential decay
- full P_rad routine added for **after** the data acquisition to access power loss quickly (same as IDL/python)
- filter register settings fixed to match sample timings in milliseconds
- improved/implemented local-storage-mode-only, so no crash at upload
- upload of error corrected voltage input to archive as well (same as IDL/Python)

CONS

- DAQ routine now more prone to crashing, hiccups and **slow processing causing sample timing errors**
- ADC bit settings not fully understood, potentially not intended setting here (wrong documentation?)
- crashing with bad device status at memory overflow or uncompleted DAQ/upload, inconsistent to recover
- hardware needs warm-up time of ~30min (before?)
- some small fixes needed to improve quality-of-life experience during experimental campaign (dials not working as intended, ...)

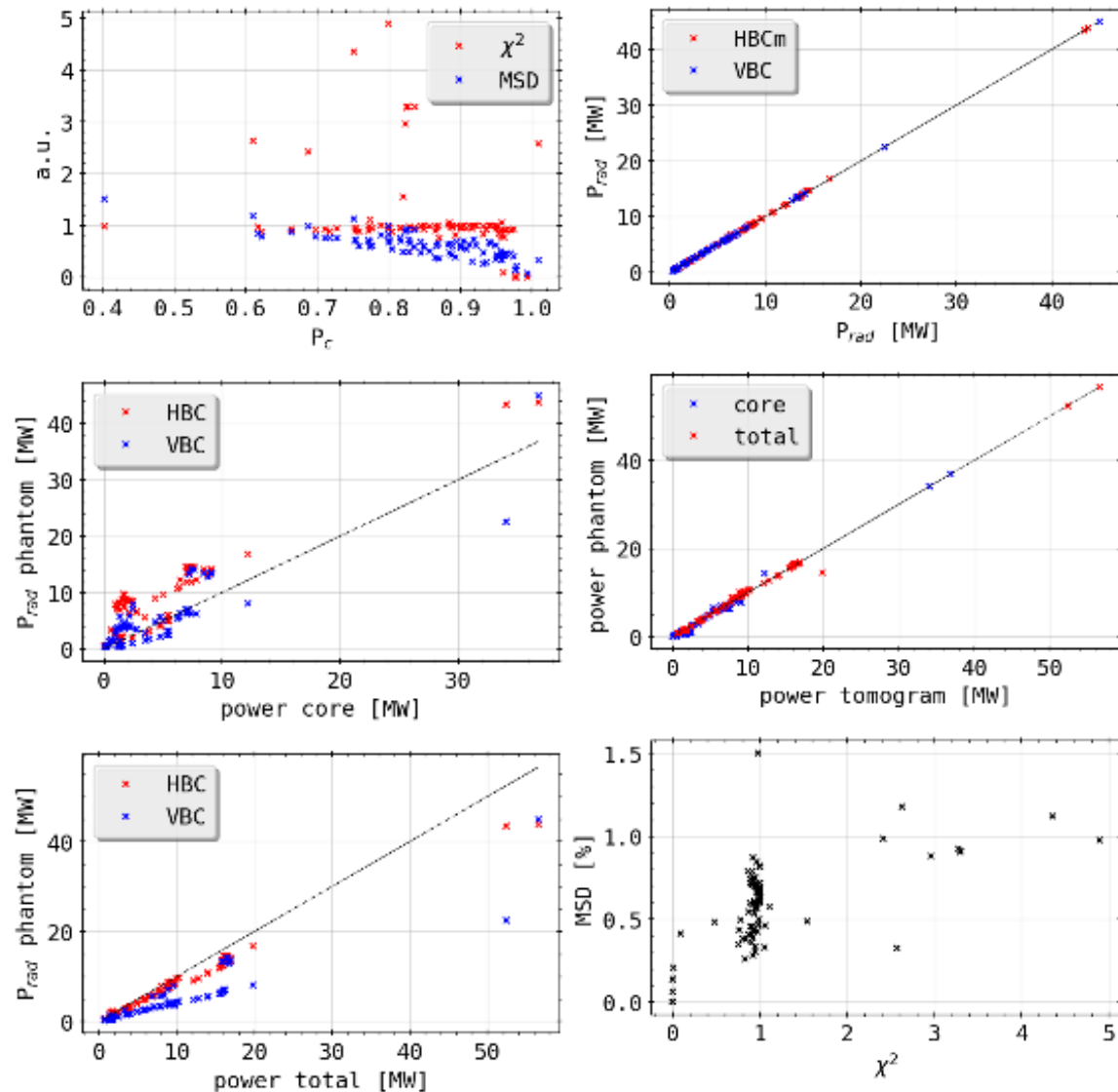
- collected MFR 2D inversion results from both phantom benchmarks and experimental data, including multiple experiments and parameter sets per target
- experimental data inversion 'limited' to comparisons of power values from either chordal (linear) profile or 2D profile of emissivity matrices
- phantom reconstruction results contain also radial profile information and error value characteristics (χ^2 , pearson coefficient, 2D deviation mean)

Tomogram Statistics: Phantom Reconstructions



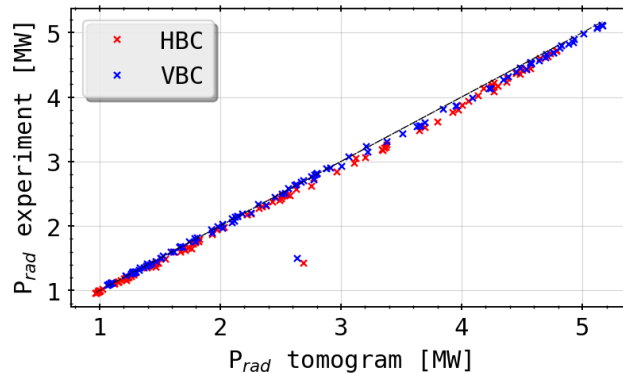
- radial profile positioning good, though width and height have small to large deviations
- peak height comparison exhibits offset on phantom side: widened radial shape and longer decays (width)
- P_{rad} and total/core power values match very well for phantom reconstructions
- a slightly larger range of MSD values yields a 'good' χ^2 of ~ 1.0
- there seems to be an offset of MSD about 0.25% (minimum error between source and target to be expected?)

Tomogram Statistics: Phantom Reconstructions



- MSD approaches 0.0 if pearson coefficient reaches 1.0, as expected, while χ^2 is around 1.0
- P_{rad} over core/total 2D integrated power yield constant offsets and slopes
- core power matches P_{rad} vor VBC(?) though is lower for HBCm as expected
- total power is larger for both P_{rad} estimates, though with different proportionalities

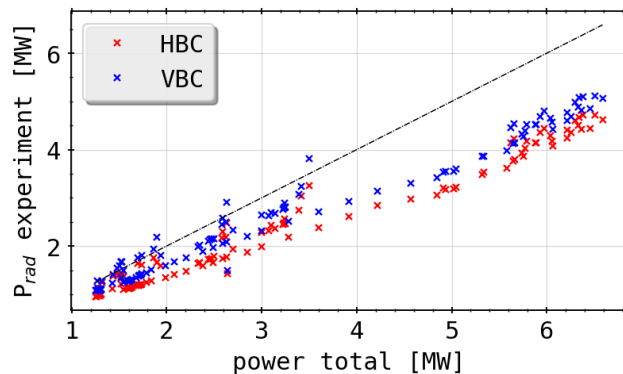
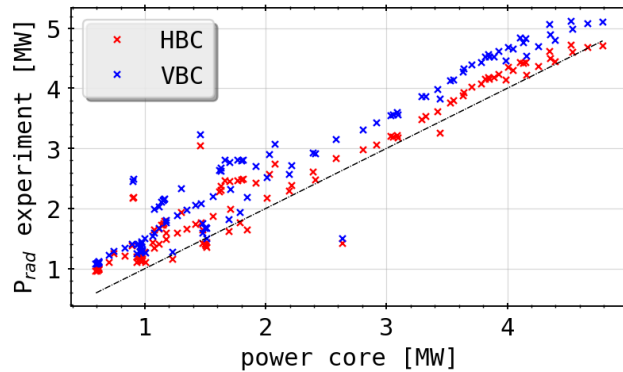
Tomogram Statistics: Experiment Reconstructions



- same experience with $P_{rad}(s)$ and total power comparison
- now absolute offset in core power comparison is closely the same for both camera arrays

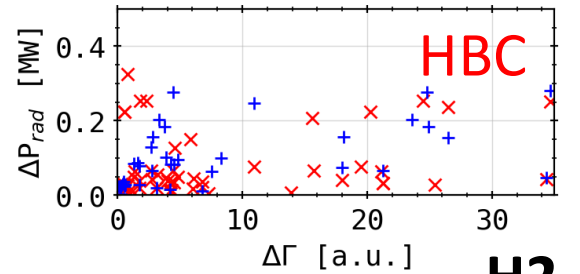
SUGGESTION

- extra slope in P_{rad} vs. total power by cutting negative results from reconstruction when limiting to only physically meaningful numbers ($P/m^3 > 0$)
- increase of their impact as expected with overall higher radiation level to counteract errors in chordal profile

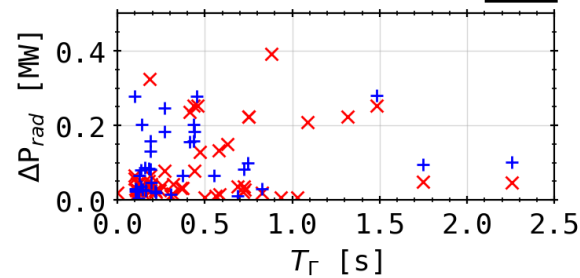


Peak Statistic: Methane Session with QSQ

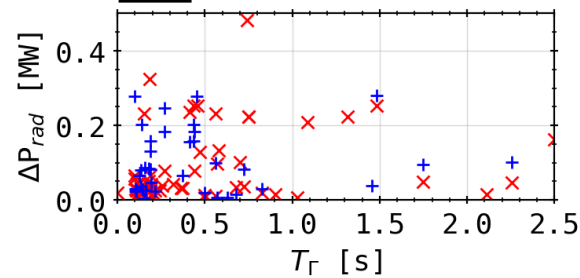
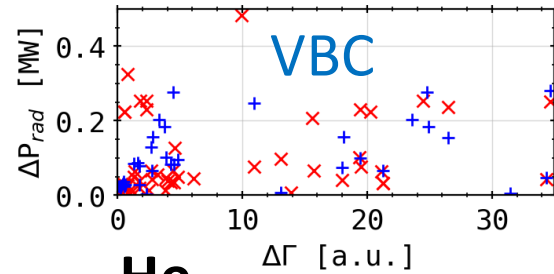
100 ms



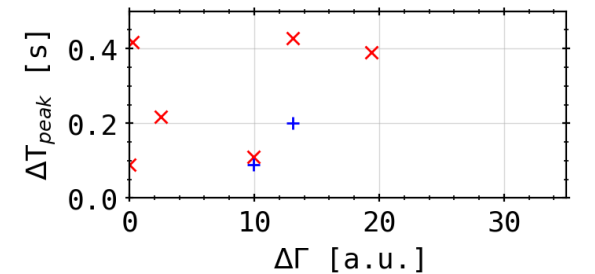
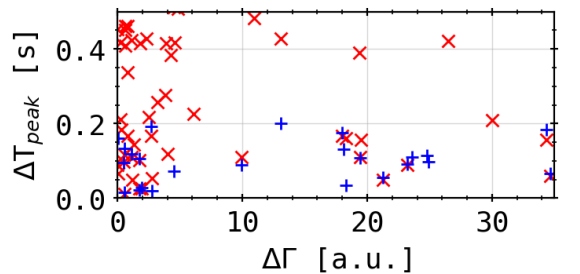
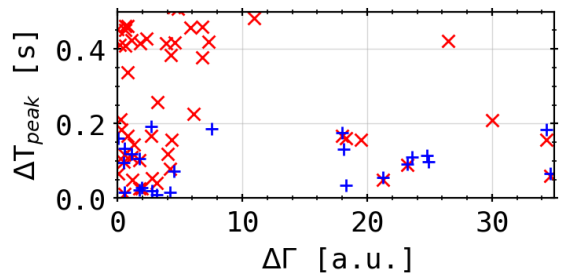
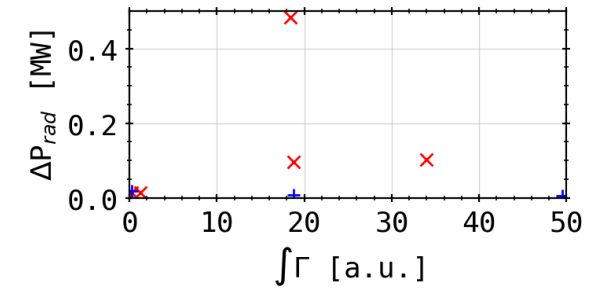
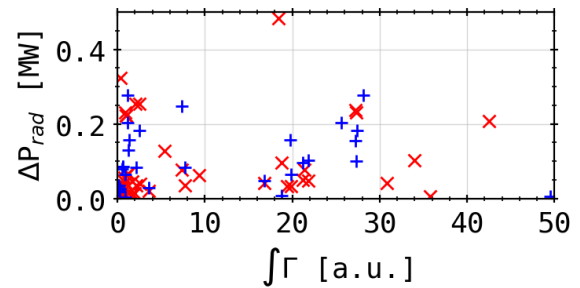
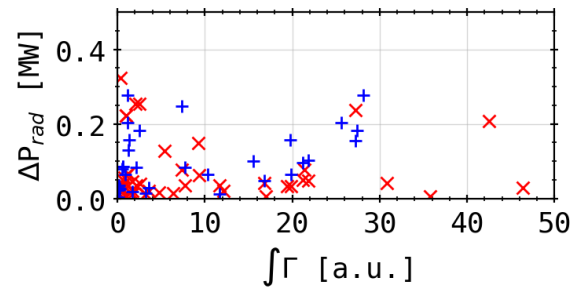
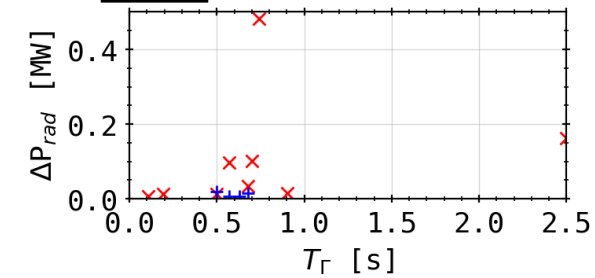
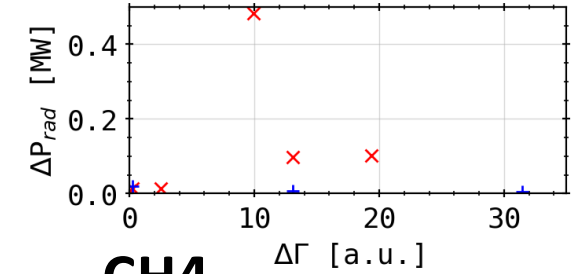
H2



He

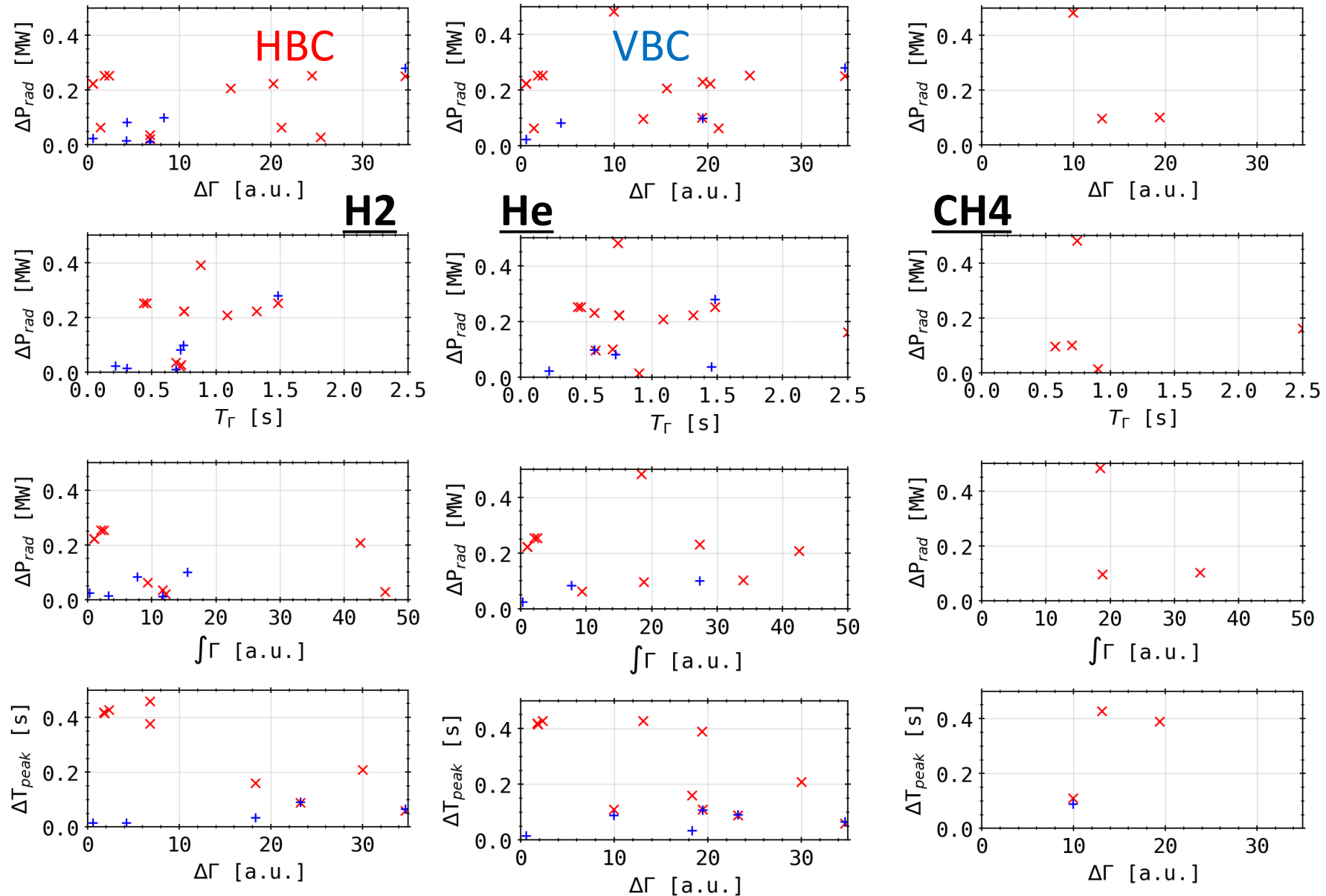


CH4



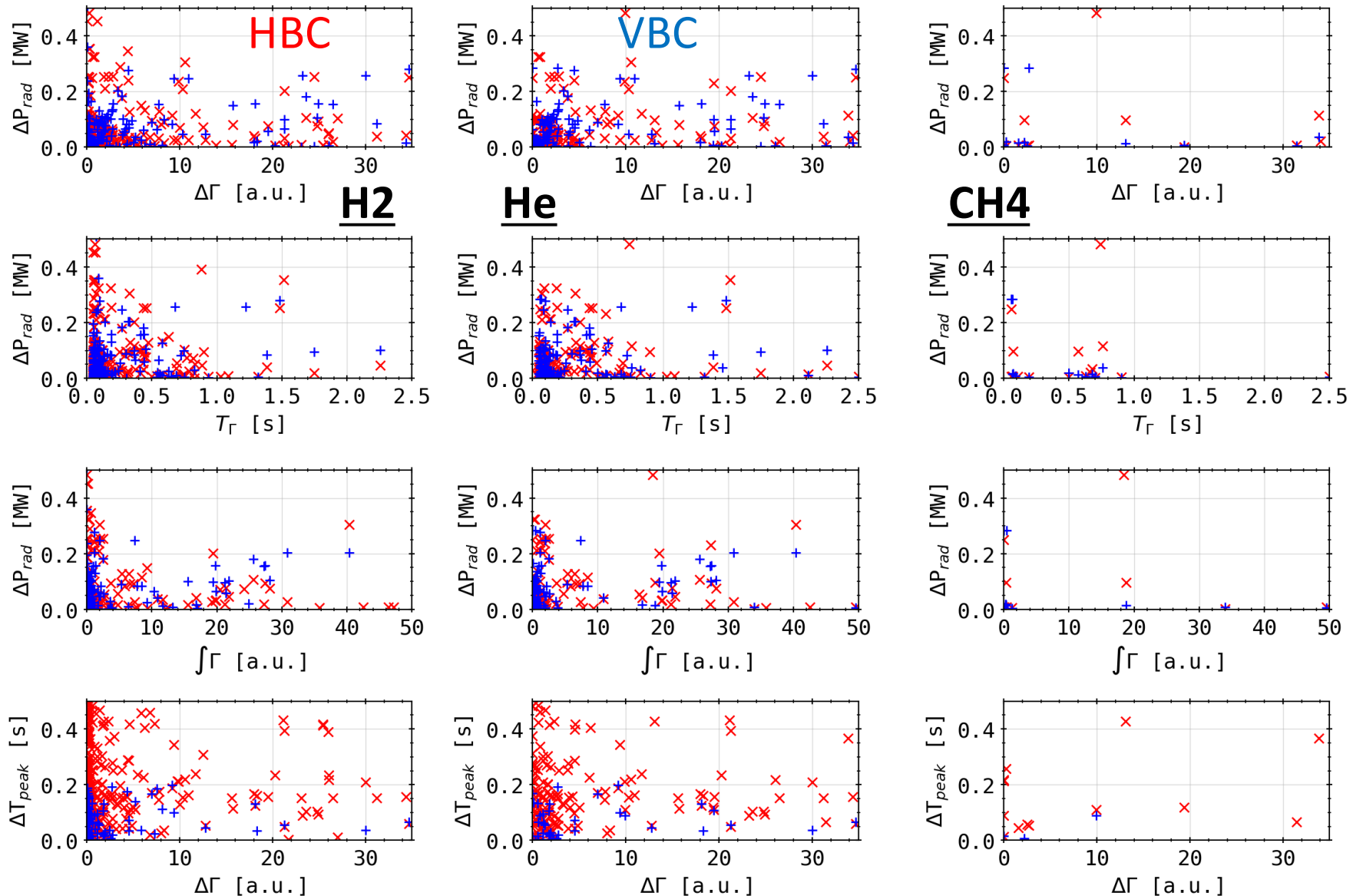
Peak Statistic: Methane Session with QSQ

200 ms



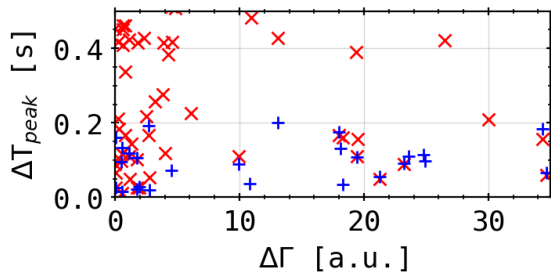
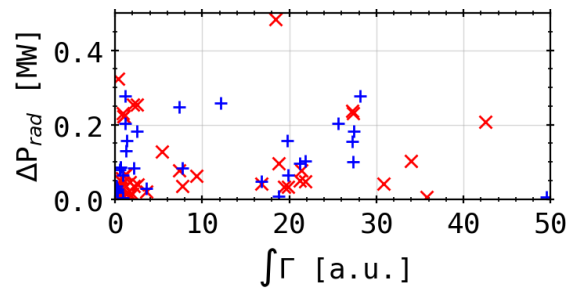
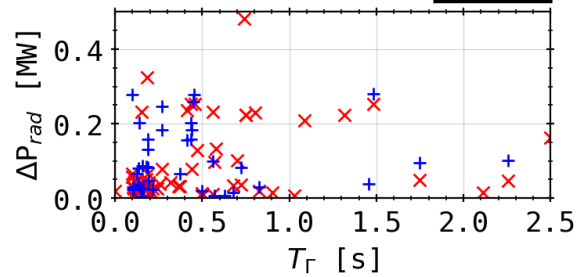
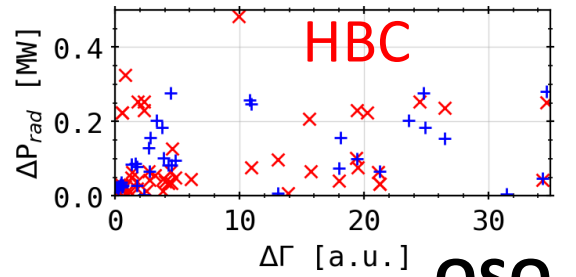
Peak Statistic: Methane Session with QSQ

50 ms



Peak Statistic: Methane Session with QSQ

100 ms



Main
Valve

